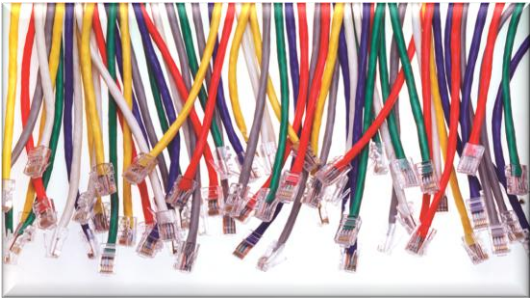


Mike Meyers' Guide to Managing and Troubleshooting Networks

Cabling and Topology



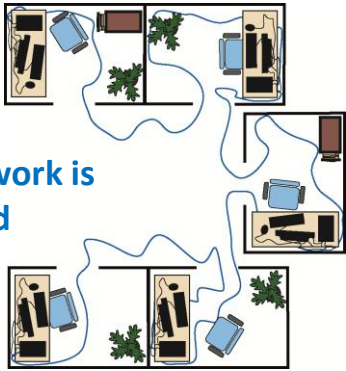
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

1

Mike Meyers' Guide to Managing and Troubleshooting Networks

Network topology: methods cables and other pieces of hardware connect to one another

- **Logical topology** – how devices connect to each other with regards to data flow
 - Also called **signaling topology**
- **Physical topology** – how a network is laid out in the messy real world
 - Also called a **schematic**



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

2



Mike Meyers' Guide to Managing and Troubleshooting Networks

Point-to-Point

- Two computers connect directly
- No need for a central hub
- Wired or wireless



Figure. Point-to-Point

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

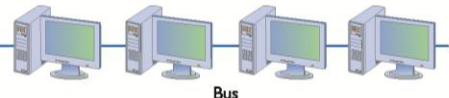
3



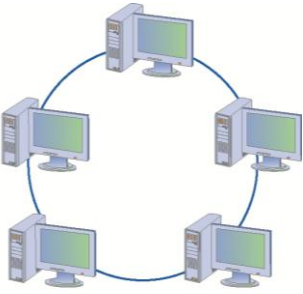
Mike Meyers' Guide to Managing and Troubleshooting Networks

Bus and Ring

- **Bus topology**
 - Single bus cable
 - Connects all computer in a line



- **Ring topology**
 - Central ring of cable
 - Connects computers in a ring



Ring

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

4

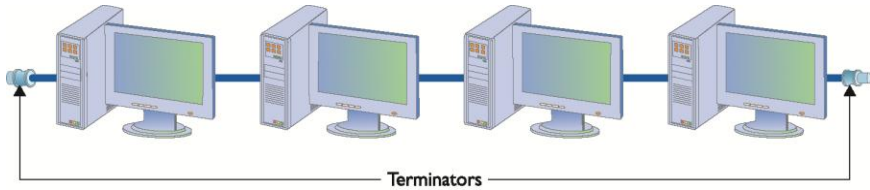


Mike Meyers' Guide to Managing and Troubleshooting Networks

Data Flow

Bus topology

- Data flows from each computer onto the bus
- Terminating resistor required at ends to prevent data reflection



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

5

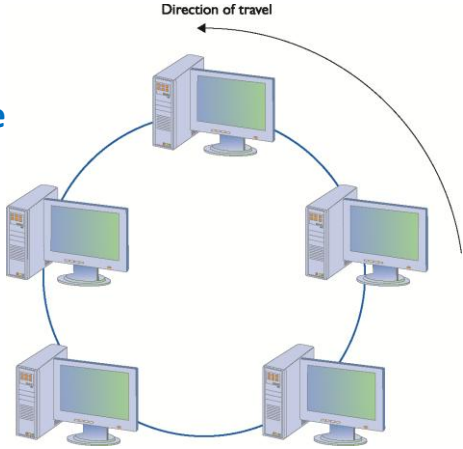


Mike Meyers' Guide to Managing and Troubleshooting Networks

Data Flow

Ring Topology

- Data flows from one computer to next one in circle
- No end of cable and no need for termination



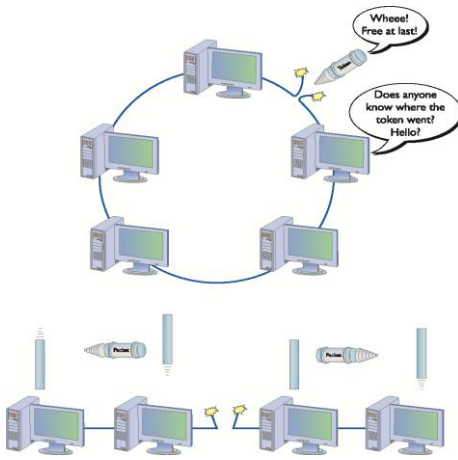
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

6



Mike Meyers' Guide to Managing and Troubleshooting Networks

Zero Fault Tolerance



Nobody is talking!

Problem with Bus & Ring

- Entire network stops working with just one break
- **Fault tolerance:** a system's capacity to keep functioning even when some part has failed

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

7

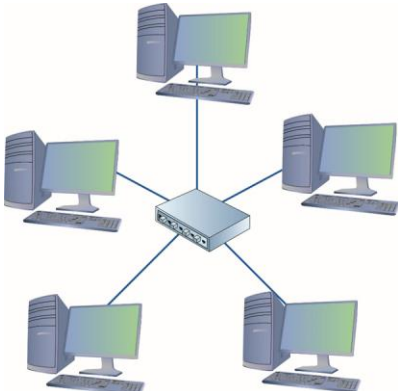


Mike Meyers' Guide to Managing and Troubleshooting Networks

Star

Star topology (hub-and-spoke)

- A central connection for all computers
- Better fault tolerance
- **Problems:**
 - More expensive
 - Difficult to redesign



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

8

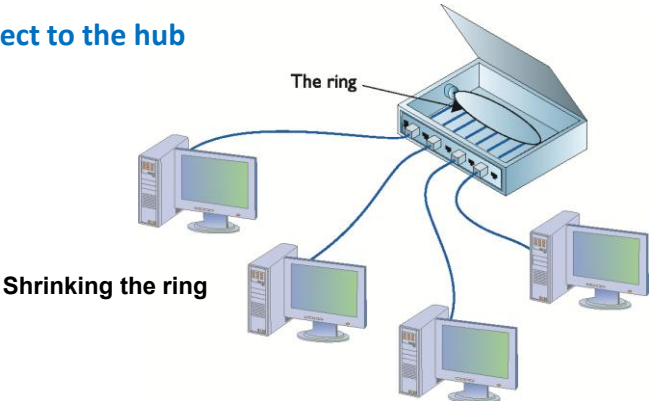


Mike Meyers' Guide to Managing
and Troubleshooting Networks

Hybrid: Star-ring

Hybrid: Star-ring topology

- Looks like a star, acts like a ring
- Ring shrunk down into a hub-like box
- Cables connect to the hub



The ring

Shrinking the ring

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

9

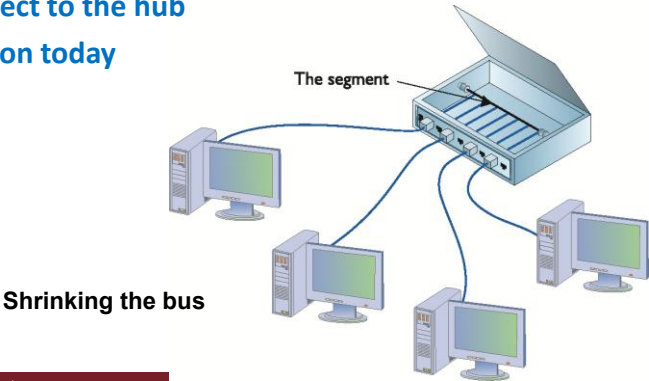


Mike Meyers' Guide to Managing
and Troubleshooting Networks

Hybrid: Star-bus

Hybrid: Star-bus topology

- Looks like a start, acts like a bus
- **Segment (bus)** shrunk down into a hub-like box
- Cables connect to the hub
- Most common today



The segment

Shrinking the bus

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

10

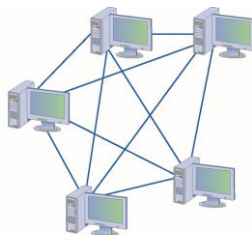


Mike Meyers' Guide to Managing and Troubleshooting Networks

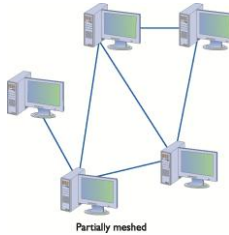
Mesh

Two types of mesh topology

- **Fully-meshed topology**
 - Every computer connects directly to every other computer
 - Most fault tolerant
 - Often used in wireless networks
 - For n devices: $\frac{n(n-1)}{2}$ links
- **Partially-meshed topology**
 - At least two machines have redundant connections



Fully meshed



Partially meshed

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

11



Mike Meyers' Guide to Managing and Troubleshooting Networks

In class Problem

- What if there are 20 nodes, how many connections in a fully meshed wired topology?

$(20-1)*20/2 = 190$

- What are two reasons why you might use a partially wired mesh topology?

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

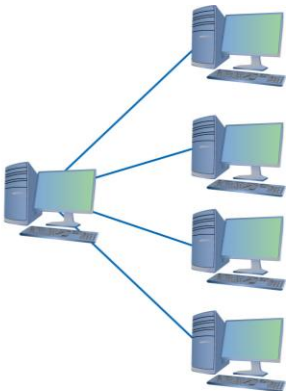
12



Mike Meyers' Guide to Managing and Troubleshooting Networks

Point-to-Multipoint

- A single system acts as a common source through which all members of the network converse
 - Requires an intelligent device in the center



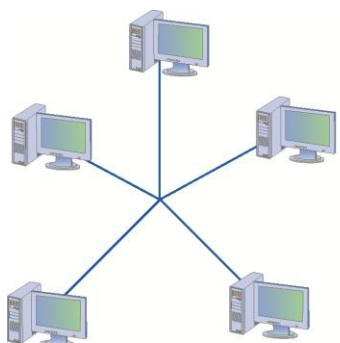
Point-to-multipoint

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

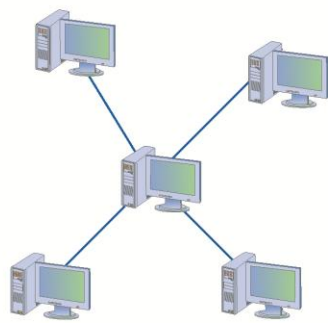
13



Mike Meyers' Guide to Managing and Troubleshooting Networks



Star



Point-to-multipoint

Comparing star and point-to-multipoint

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

14



Mike Meyers' Guide to Managing and Troubleshooting Networks

Pros & Cons

Bus Physical Topology

- **Pros:**
 - Easy to install
 - Uses less cabling than other topologies
- **Cons**
 - Electrical limits on length of bus, number of taps
 - Not fault tolerant – a break in the backbone usually disables all hosts on the network
 - Every host on the network can see all traffic

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

15



Mike Meyers' Guide to Managing and Troubleshooting Networks

Pros & Cons

Ring Physical Topology

- **Pros:**
 - Easy to install and reconfigure
- **Cons:**
 - Not fault tolerant – a break in the ring usually disables all hosts on the network
 - Every host on the network can see all traffic

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

16



Mike Meyers' Guide to Managing and Troubleshooting Networks

Pros & Cons

Star Physical Topology

- **Pros:**
 - Easy to install and reconfigure
 - Robust – a link failure affects only one computer
- **Cons:**
 - May require more cabling than a bus or a ring
 - Connection point is a single point of failure
 - Every host on the network can usually see all traffic

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

17



Mike Meyers' Guide to Managing and Troubleshooting Networks

Pros & Cons

Full Mesh Physical Topology

- **Pros:**
 - Most robust
 - Dedicated links, so no traffic problems due to shared links
 - Private and secure links
- **Cons:**
 - Expensive due to the number of ports and number of links required, but can be done wirelessly

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

18



Mike Meyers' Guide to Managing and Troubleshooting Networks

Pros & Cons

Point-to-Point Physical Topology

- **Pros:**
 - Very simple
 - No extra hardware, such as a central hub, is required
 - Private and secure link
- **Cons:**
 - Cannot expand the network

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

19



Mike Meyers' Guide to Managing and Troubleshooting Networks

Pros & Cons

Point-to-Multipoint Physical Topology

- **Pros:**
 - Need a host to act as the hub
 - Private and secure links
- **Cons:**
 - The capacity of the hub host limits the size of the network

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

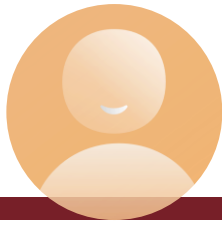
20



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Cabling

Coaxial Cable Twisted Pair Fiber



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

21



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Parameters of a topology

- Topology is only one feature of a network
- Other network features
 - What is the cable made of?
 - How long can it be?
 - How do machines decide which machine should send data and when?

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

22



Mike Meyers' Guide to Managing and Troubleshooting Networks

Bandwidth

- In signal processing,
 - **Range of frequencies a medium can pass without losing half the power**
 - Ex: A cable can pass frequencies between 10 KHz and 50 KHz

$$\text{BW} = \text{high freq} - \text{low freq}$$

$$\text{BW} = 50 - 10 = 40 \text{ KHz}$$
- In computing,
 - **Number of bits per second (bps) that can be transmitted**
 - **Bits/second is proportional to bandwidth in hertz**
 - **Networks use bandwidth-efficient encoding schemes**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

23



Mike Meyers' Guide to Managing and Troubleshooting Networks

Bandwidth

Assume a file is 10 Mbytes long. How long will it take to transmit the file over a 100Mbps link?

$$10 \text{ Mbytes} \times 8 \text{ bits/byte} = 80 \text{ Mbits}$$

$$80 \text{ Mbits} \times \frac{1 \text{ sec}}{100 \text{ Mbits}} = 0.8 \text{ seconds}$$

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

24



Mike Meyers' Guide to Managing and Troubleshooting Networks

In class Problem

How long will it take to transmit a 10MB file over a 1Gbps link?

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

25



Mike Meyers' Guide to Managing and Troubleshooting Networks

In class Problem

How long will it take to transmit a 10MB file over a 1Gbps link?

$$\frac{10 \text{ MB}}{1 \text{ file}} * \frac{8 \text{ b}}{1 \text{ B}} * \frac{10^6 \text{ B}}{1 \text{ MB}} * \frac{1 \text{ sec}}{1 \text{ Gb}} * \frac{1 \text{ Gb}}{10^9 \text{ b}}$$

$$= 8 * 10^{-2} \text{ seconds}$$

$$= 0.08$$

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

26



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Bandwidth

Different kinds of cables have different bandwidth

- **Twisted Pair**
 - Bandwidth varies with different kinds of cables
- **Coaxial Cable**
 - Much higher bandwidth than twisted pair
 - Carries higher-frequency signals than twisted pair
- **Fiber**
 - Much higher bandwidth than coaxial or twisted pair

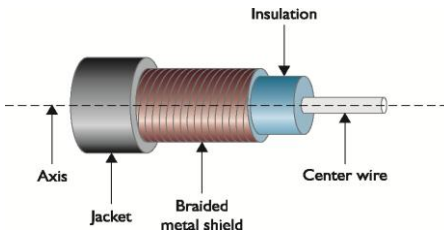
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

27



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Coaxial Cable



Cutaway view of coaxial cable



Coaxial cable showing braided metal shielding

Central conductor wire, surrounded by

- **Insulating material then a braided metal shield**
 - Braided metal shield layer of coaxial cable
- **Shields transmissions from electromagnetic interference (EMI)**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

28

Mike Meyers' Guide to Managing and Troubleshooting Networks

Coaxial Cable

- Used in older computer networks

T shaped BNC

Coax cable

Network Interface Card (NIC) with BNC

- Coaxial cable is used today primarily to enable cable modems, TV cable boxes, and video connections, such as antennas or satellite receivers

F-type connector on coaxial cable

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

29

Mike Meyers' Guide to Managing and Troubleshooting Networks

Coaxial Cable

- RG rating for coaxial cable
 - **RG-6 is dominate cable today**
 - **RG-58 and RG-59 cable rarely used**
- Coaxial cable Ohm rating
 - **Relative measure of impedance**
 - **RG-6 and RG-59 are rated at 75 Ohms**
 - **Make sure to use same-rated cables within a network**

RG-6 cable

Ohm rating (on an older, RG-58 cable used for networking)

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

30



Mike Meyers' Guide to Managing and Troubleshooting Networks

Coaxial Cable

- Splitting coaxial cable
- Extending coaxial cable



Coaxial splitter



Barrel connector

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

31



Mike Meyers' Guide to Managing and Troubleshooting Networks

Twisted Pair

- Most common network cabling
- Twisted pairs of cables, bundled together
- Twists reduce **crosstalk** interference
- Two Types
 - Shielded twisted pair (STP)
 - Unshielded twisted pair (UTP)

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

32



Mike Meyers' Guide to Managing and Troubleshooting Networks

Shielded Twisted Pair

- Shielding protects from electromagnetic interference (EMI)
- Needed in locations with excessive EMI
- Most common is IBM Type 1 cable



Shielded twisted pair (STP)

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

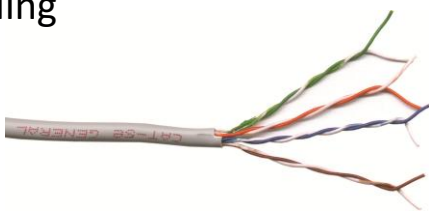
33



Mike Meyers' Guide to Managing and Troubleshooting Networks

Unshielded Twisted Pair

- Most common
- Twisted pairs of wires with plastic jacket
- Cheaper than STP
- Also used in telephone systems, Ethernet (LAN) cabling



Unshielded twisted pair (UTP)

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

34

Twisted Pair

Copyright © 2022 by McGraw-Hill Education. All rights reserved.



Using the Correct Cable

-

A close-up photograph of a yellow Ethernet patch cable. A yellow callout box with a black border and a pointer indicates the text "CAT level" on the cable's jacket. The cable itself has black text printed on it, including "VENTS CATEGORY 5e PATCH CABLE" and "BELKIN COMPONENTS CAT". The Belkin logo is also visible.

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

18

Mike Meyers' Guide to Managing and Troubleshooting NetworksRegister Jack Connector

- **RJ-11** (two pairs of wires) for telephones
- **RJ-45** (four pairs of wires) for networks


RJ-11 (top) and 8P8C/"RJ-45" (bottom) connectors


Copyright © 2022 by McGraw-Hill Education. All rights reserved.

37

Mike Meyers' Guide to Managing and Troubleshooting NetworksFiber-Optic

- **Fiber-optic cable** transmits light
- Not affected by EMI
- Excellent for long-distance transmissions
 - **Single copper cable works up to a few hundred meters**
 - **Single fiber-optic cable works up to tens of kilometers**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

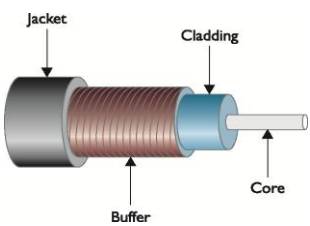
38



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Fiber-optic Composition

- **Core:** the glass fiber
- **Cladding:** reflects signal down the fiber
- **Buffer:** gives strength
- **Insulating jacket:** protects inner components
- Two-number designator
 - **Core / cladding measurements**
 - 62.5/125 μm – the most common cable size



Cross section of fiber-optic cabling

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

39



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Fiber-optic

- Almost all network technologies use fiber optic cable with pairs of fibers
 - One for sending
 - One for receiving
- Two types
 - Multimode fiber (MMF)
 - Single-mode fiber (SMF)



Duplex fiber-optic cabling

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

40



Mike Meyers' Guide to Managing and Troubleshooting Networks

Light Sources

- Two possible light sources
 - **Light Emitting Diodes (LEDs)** – called **multimode**
 - Usually, 850 nm wavelength
 - **Lasers** – called **single-mode**
 - Prevents **modal distortion** (a problem with multimode)
 - High transfer rates over long distances
 - 1310 or 1550 nm wavelength

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

41



Mike Meyers' Guide to Managing and Troubleshooting Networks

LEDs versus Lasers

• LEDs – Low cost – Slower – Easy to use • Laser – Higher cost – Faster – Harder to use	• Examples – Printers - LEDs • Oki Data Americas – Headlamps – lasers • BMW to build laser lamps – Networks • Onsite cabling uses LEDs more often • Long distance telephony uses Laser
--	--

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

42



Mike Meyers' Guide to Managing and Troubleshooting Networks

Connectors

- ST: bayonet-style
- SC: push-in
- LC: duplex
- MT-RJ



MT-RJ fiber-optic connector



From left to right: ST, SC, and LC fiber-optic connectors

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

43



Mike Meyers' Guide to Managing and Troubleshooting Networks

In class discussion

Give some pros/cons between using UTP vs fiber optic cabling?

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

44



Mike Meyers' Guide to Managing and Troubleshooting Networks

Cable Fire Ratings

Types of Ratings – controlled by code

- **Polyvinyl chloride (PVC)-rated cable**
 - Rating has no significant fire protection
 - Lots of smoke and noxious fumes
- **Plenum-rated cable**
 - Costs three to five times more than PVC cable
 - Often required for all cabling
 - Burns with much less smoke and fumes
 - Most city ordinance require plenum cable for installations
- **Riser-rated cable**
 - Used for cables run in walls (or vertically)
 - Less protection than plenum cable
 - Most installations today use plenum for runs between floors

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

45



Mike Meyers' Guide to Managing and Troubleshooting Networks

Connecting Devices

- Repeater
 - Operates at the physical layer of the layered model
 - Connects two segments of the same LAN
 - Regenerates signals travelling on the LAN to extend the maximum length
 - Forwards everything
- Hub
 - Physical layer device
 - Multiport repeater
 - Sends frames received on one port out to all the other ports

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

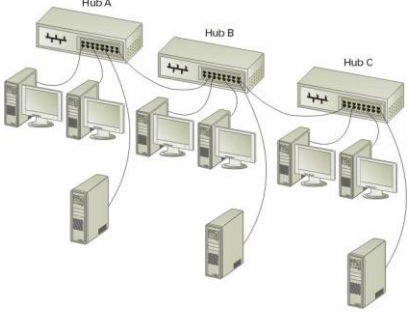
46



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Connecting Devices

- Adding Hubs
 - May expand the network by connection multiple hubs together



- As more nodes are added to the network traffic increases
- Too many hubs can reduce network performance

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

47



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Connecting Devices

Using Switches

- **Switch (also called a bridge or a layer 2 switch)**
- **Operates at physical layer and the data link layers**
 - Physical layer: acts as a hub
 - Data link layer: examines the physical (MAC) addresses in the frame to send frames received on one port to only the destination port
- **Used to connect LANs in the same organization and location together**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

48

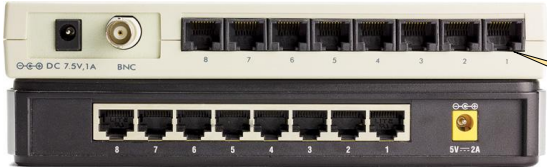


Mike Meyers' Guide to Managing
and Troubleshooting Networks

Connecting Devices

Switches and Hubs

- Switches used to be far more expensive than hubs, but today they are about the same price
- Switches have become the standard network appliance – replacing hubs



Top is a hub
Bottom is a switch

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

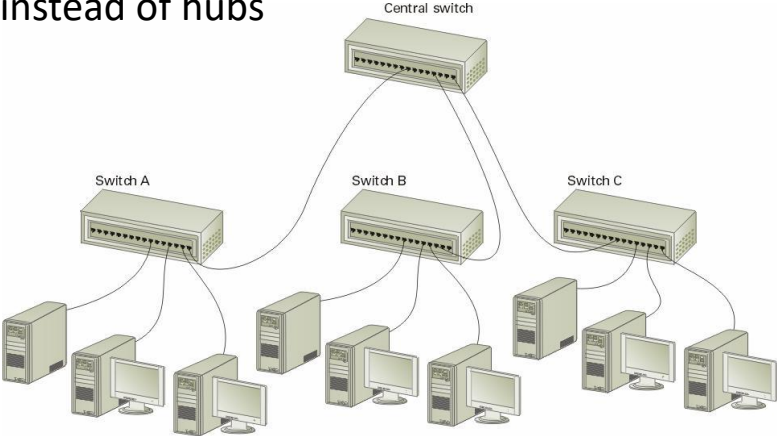
49



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Switches

- A network with three switches installed instead of hubs



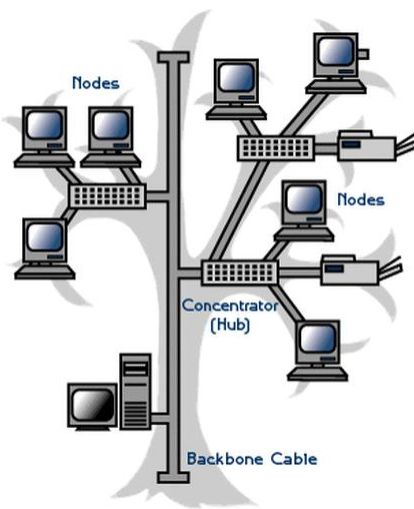
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

50



Mike Meyers' Guide to Managing and Troubleshooting Networks

Tree Topology



- Pros
 - More fault tolerant
 - Used in broadband
 - Expandable
 - Manageable
 - Error detection
 - Point-to-point wiring
- Cons
 - Troubleshooting more complex
 - Costly
 - Backbone can be a single point of failure
 - Reconfiguring can be a challenge (wiring)

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

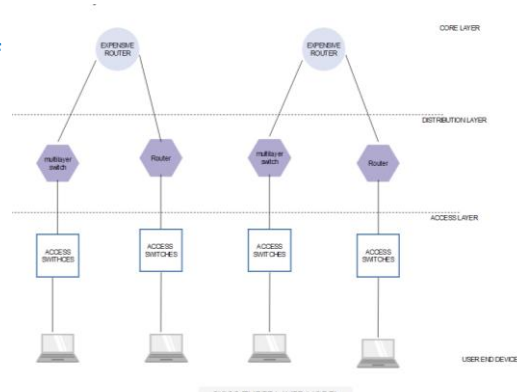
51



Mike Meyers' Guide to Managing and Troubleshooting Networks

Cisco 3 Layer Model

- Core Layer
 - Manages large amounts of high-speed traffic
 - Backbone
- Distribution Layer
 - Security/firewalls
 - Performs WAN filtering
- Access Layer
 - User interact with devices on the network



CISCO THREE LAYER MODEL

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

52



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Installing a Physical Network



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

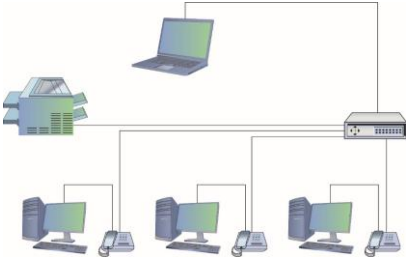
53



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Structured Cabling

- Governed by a set of TIA/EIA standards
 - **Telecommunications Industry Association/Electronic Industry Alliance**
- Details on every aspect of cabled network
 - **Assessment of connections leading outside the network**
 - **Type of cabling**
 - **Position of wall outlets**
 - **Network components**



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

54



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Structured Cabling

- Goal
 - Create a safe, reliable cabling infrastructure
 - Communicate knowledgeably with cable installers & perform basic troubleshooting
- Applies to
 - Networks
 - Telephone
 - Video conferencing
 - Anything that needs low-power, distributed cabling

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

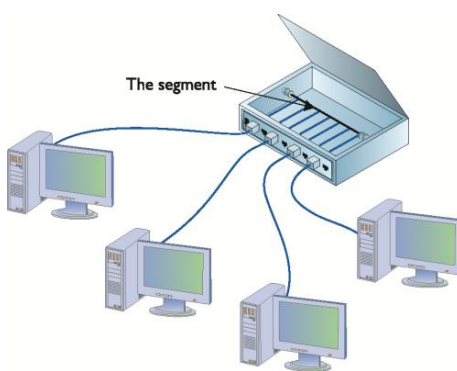
55



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Basic Network

- Most common network: star- bus topology
 - A switch - either a layer 2 switch (also called a **bridge**) or a hub (layer 1 device)
 - UTP cable
 - 2 or more PCs



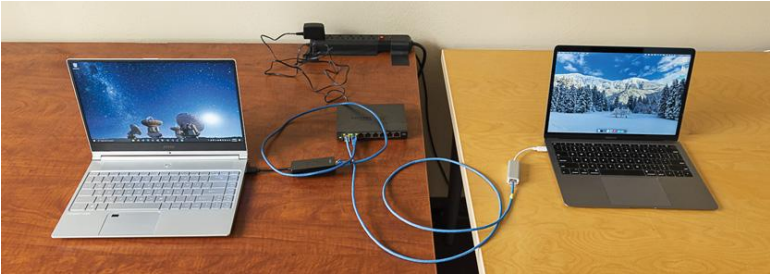
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

56

Mike Meyers' Guide to Managing and Troubleshooting Networks

Basic Network

A switch connected by UTP cable to two PCs



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

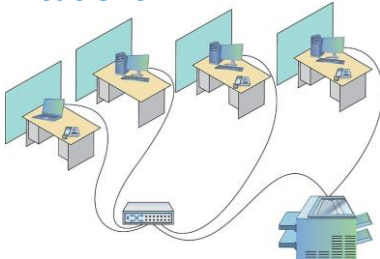
57

Mike Meyers' Guide to Managing and Troubleshooting Networks

Switch location

Problems to consider

- **People tripping over cables**
- **EMI**
- **Exposed cables vulnerable to physical damage**
- **Challenges to network modifications**



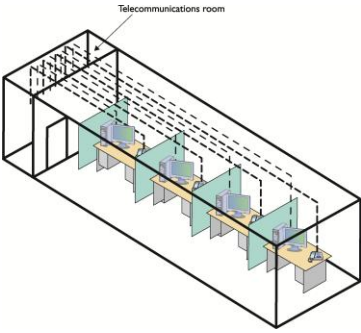
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

58



Mike Meyers' Guide to Managing and Troubleshooting Networks

Structured Cabling



Telecommunications room

A Better Approach

- Keep switches in a telecommunications room (sometimes called a server room)
- Use hardware in the room to organize and protect the cabling

A Better Installation Design

- Provides safety
- Provides hardware to organize and protect cabling
- Uses a cabling standard with flexibility to allow for growth
 - TIA/EIA developed standards for cable installation

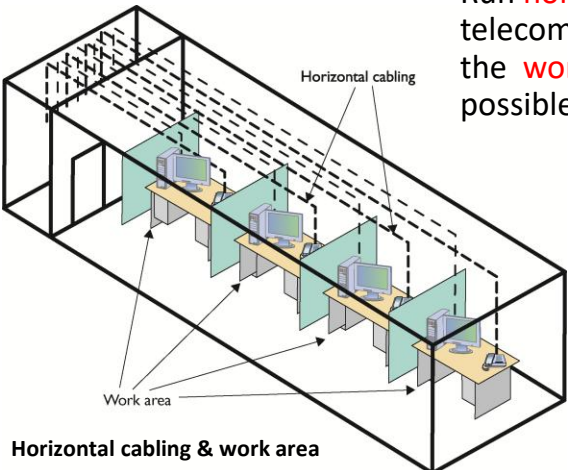
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

59



Mike Meyers' Guide to Managing and Troubleshooting Networks

Structured Cabling



Horizontal cabling

Work area

Run horizontal cable from the telecommunications room to the work area in the ceiling if possible

Horizontal cabling & work area

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

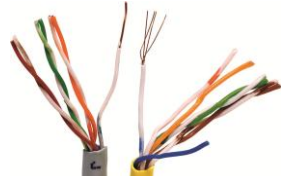
60



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Horizontal Cabling

- Runs from work area to telecommunications room
- Use CAT5e or better UTP cable
 - CAT6a for 10GBaseT
- Two types of cabling
 - Solid core - better conductor, but stiff
 - Stranded core - not as good a conductor, but does not break as easily

- Horizontal cabling – number of strands
 - Four-pair UTP assumed
 - High-end telephone setups use 25- or 100-pair
 - TIA/EIA specifies horizontal cabling should always be solid core

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

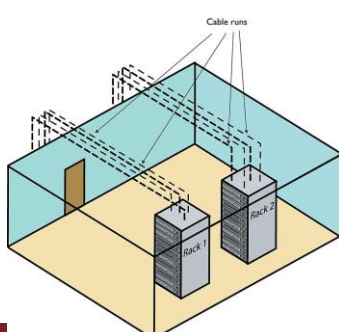
61



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Telecommunications Room

- Heart of basic star topology and one endpoint of the horizontal runs
- Central component is **equipment rack**
 - Safe, stable platform for hardware components
 - Almost all network equipment can be rack-mounted



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

62



Mike Meyers' Guide to Managing and Troubleshooting Networks

Floor-to-ceiling Rack

Short equipment rack



Rack mounted UPS



- Types of racks
 - Two post
 - Four post
 - Server rail rack
- Rack Equipment
 - 19 inches wide
 - Height measured in U, equivalent to 1.75 inches
 - Ex: 60U rack is 105 inches high
- Racks hold
 - Servers
 - UPS, PDU (power distribution unit)
 - Routers, switches




Copyright © 2022 by McGraw-Hill Education. All rights reserved.

63



Mike Meyers' Guide to Managing and Troubleshooting Networks

Patch Panels and Cables

- Don't want to connect horizontal cabling directly to the switch. Instead, use a **patch panel**
 - Row of female connectors (ports) in front
 - Permanent connections to horizontal cables in back called a **punchdown block**
 - Connect cables with punchdown tool
 - 110 block
- Proper labeling is crucial to organization of cables



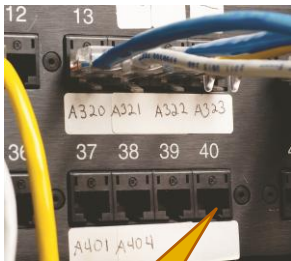
Copyright © 2022 by McGraw-Hill Education. All rights reserved.

64

Mike Meyers' Guide to Managing and Troubleshooting Networks

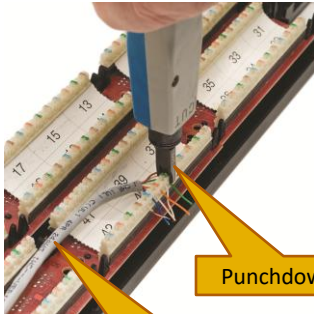
Patch Panel & Cables

Typical patch panel



Female connector

Punching down a 110 block



Punchdown tool

Solid core horizontal cable

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

65

Mike Meyers' Guide to Managing and Troubleshooting Networks

Patch Panel & Cables

- Connect patch cable to front of the patch panel and then to switch
 - Match UTP patch panel ratings with UTP cable rating
- Wide variety of panel types
 - Cable/fiber, number of ports, etc.
- Make sure to label everything
 - Use a single labeling scheme




Copyright © 2022 by McGraw-Hill Education. All rights reserved.

66



Mike Meyers' Guide to Managing and Troubleshooting Networks

Patch Panel & Cables

Typical patch cable



- Typical length UTP
 - 2 to 5 feet but 5 meters is a recommended maximum
- Stranded cable
- Cable Management
 - Color-coded
 - Document everything clearly & carefully

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

67



Mike Meyers' Guide to Managing and Troubleshooting Networks

Alternative Cable Connection Points

- BIX block by Nortel Networks
 - Building Industry Cross-connect
 - Installed on a wall rather than in a rack
- Krone block by the Krone Group
 - Krone connectors enable networking as well as audio interconnections



Krone block

BIX block

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

68

Mike Meyers' Guide to Managing and Troubleshooting Networks

Work Area

- Wall outlet
 - Female jack with CAT rating to accept cabling
 - Mounting plate, and faceplate
- Connect **NIC (Network Interface Card)** in the computer to wall outlet with patch cable
- Work area is source of most network failures



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

69

Mike Meyers' Guide to Managing and Troubleshooting Networks

Cable Length

- Most Ethernets specify a maximum length of 100 meters between the switch and each computer
- In practice, make sure runs are < 90 m
 - **Allows up to 5m of patch cable in the telecommunications room**
 - **Allows up to 5m of patch cable in the work area**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

70



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Building - Wide Networks

- Cabling on one floor a star topology
- High-speed backbone runs vertically
 - Backbone connects to multispeed switch on each floor
 - Dedicated telephone cabling backbone runs alongside
- Demarc** (demarcation point) is the connection from the outside world to the building's network
 - Dividing line of responsibility
 - Stores telephone & LANS cross-connects
- Network interface unit (**NIU**)
 - Serves as a demarc between a home network and ISP
 - DSL or cable modem usually serves as a demarc in a home



network interface box

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

71



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Demarc and NIU



Typical office demarc

- Demarcation point (Demarc):** Dividing line of responsibility
 - Physical location of the connection with the outside world
 - Marks the dividing line of responsibility for network function
- Network Interface Unit (NIU):**
 - Serves as the demarc between a Small Office Home Office (SOHO)



Typical NIC

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

72



Mike Meyers' Guide to Managing
and Troubleshooting Networks

**Connections Inside
Demarc**

- Challenge to ISP/telephone provider
 - **The need to diagnose system faults**
- Smart jacks
 - **NIUs with remote loopback capability**
- Connections inside the demarc
 - **Network and telephone cables connect to customer-premises equipment (CPE)**
- Cabling runs from NIU to CPE
 - **Might be a multiplexer or a LAN switch connected to a patch panel**
- Main patch panel is called a **vertical cross-connect**



LAN vertical cross-connect

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

73



Mike Meyers' Guide to Managing
and Troubleshooting Networks

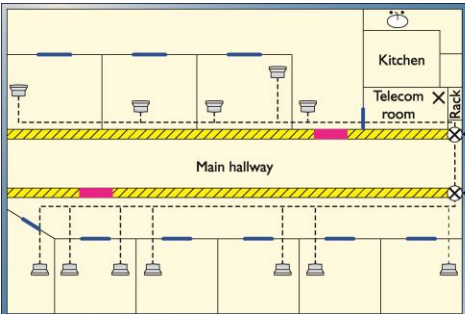
Installing Structured Cabling

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

74

Mike Meyers' Guide to Managing and Troubleshooting Networks

Installing Structure Cabling



Hand-drawn network floor plan`

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

Getting a Floor Plan - Key to proper planning

- **Determine potential locations for telecommunications rooms**
- **Locate physical firewalls**
- **Gives an overall feel for scope of job**
- **If no floor plan, create one**

75

Mike Meyers' Guide to Managing and Troubleshooting Networks

Installing Structured Cabling

Mapping the Runs

- **Determine length of cable runs**
 - None longer than 90 meters
- **Determine route of cable runs**
 - Keep cables together as long as possible in hooks or cable trays or outside the wall in a raceway
- **Determine location of each cable drop**
 - Where cable comes out of the wall
 - Include extra drops
- **Determine customer's plans & budget**
 - Most network installers quote a per-drop cost
- **Decide if cable runs will be inside or outside walls**
 - External raceway products



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

76

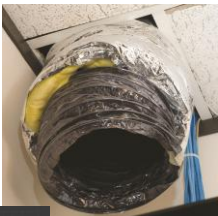
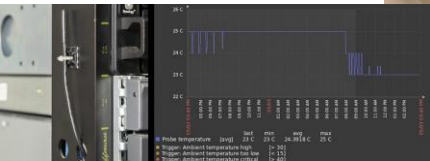


Mike Meyers' Guide to Managing
and Troubleshooting Networks

Tecom Location

Criteria to consider

- **Distance** no more than 90 meters from drops
- **Make sure there is more than enough power**
 - Put the room on its own dedicated circuit
- **Low humidity**
- **Provide cooling**
- **Make sure there is room to expand**
- **Door should be lock-able to control access**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

77



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Installing Structured Cabling

Pulling Cable

- **Start in telecommunications room and pull to drops**
- **Open drop ceiling and string via hooks or cable trays**
- **Requires 2 to 3 people**
- **Hardest part: working around old cable installations in the ceiling**
- **Neatness counts!**



Cable trays over a drop ceiling

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

78



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Installing Structured
Cabling

Messy cabling nightmare



Good cable management



Copyright © 2022 by McGraw-Hill Education. All rights reserved.



Mike Meyers' Guide to Managing
and Troubleshooting Networks

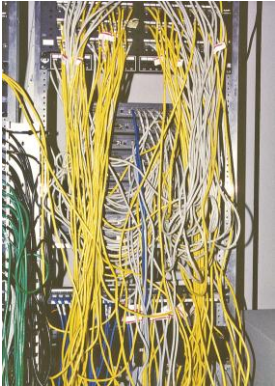
Installing Structured
Cabling

End of cables guided to rack

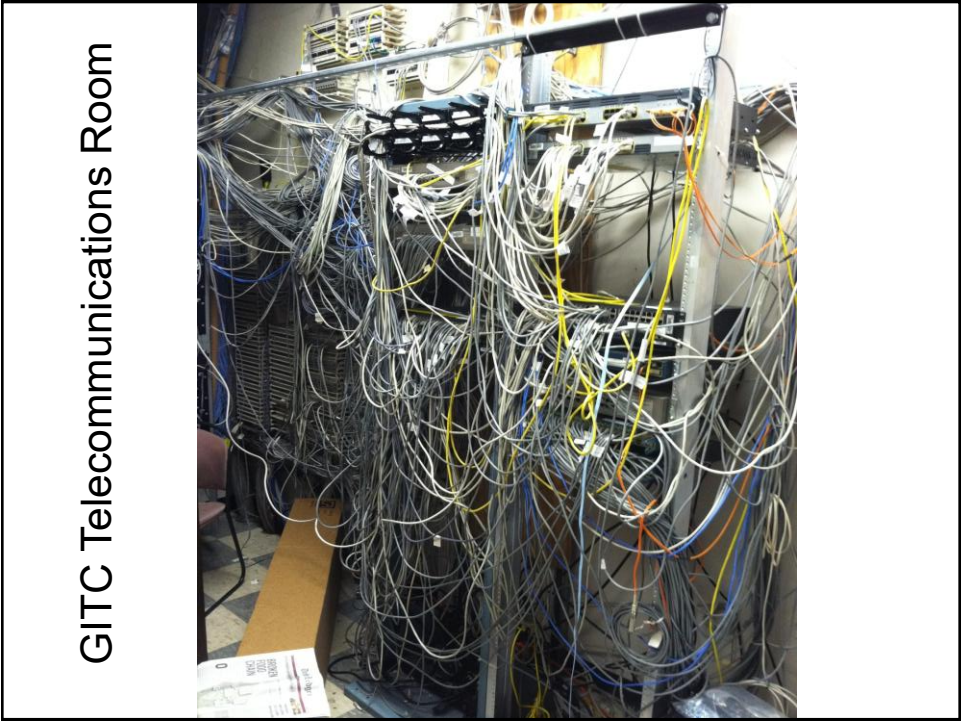


Cable guides

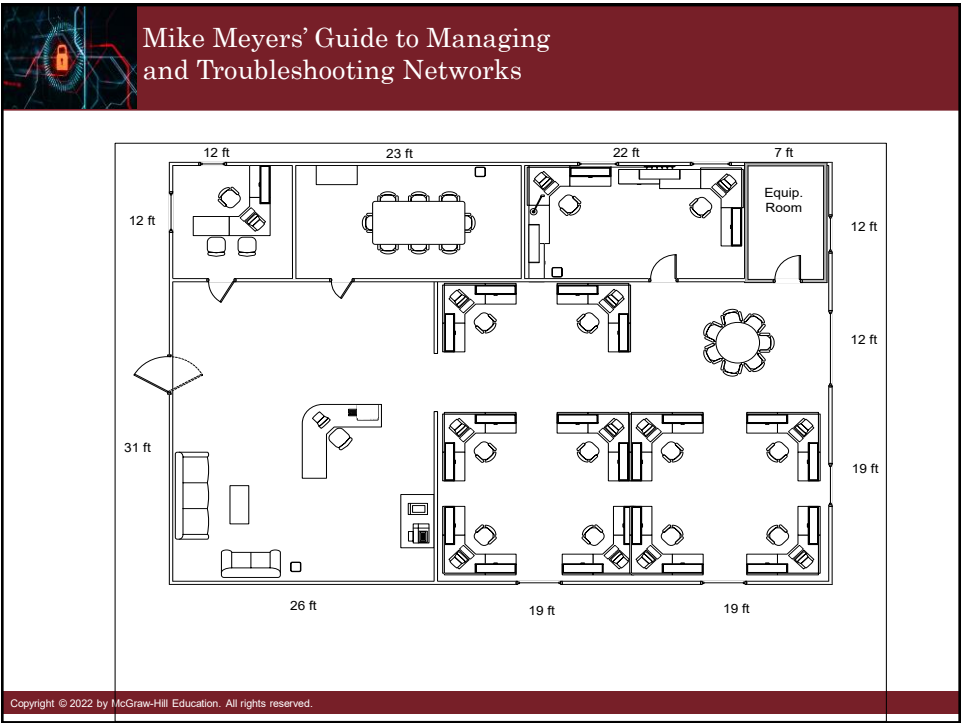
Bad cable management



Copyright © 2022 by McGraw-Hill Education. All rights reserved.



81



82



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Testing Cable Runs

- Verify each jack is properly connected
- Verify cable run can handle the network speed
 - **Copper and fiber have different challenges**
 - **Consider...**
 - Signal degradation
 - Lack of connection
 - Interference
- Testing tools
 - **Enable you to diagnose network problems**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

83



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Testing Cable Runs

Copper Challenges

- **Potentially bad copper cable**
 - Incorrect length or speed
 - Broken or shorted wires
 - Incorrectly terminated when connected
 - Electrical or radio interference
 - Crosstalk and attenuation
- **Testing tools**
 - Cable tester – verifies cable and terminated ends are correct
 - Continuity tester – run a wiremap test
 - Multimeter
 - Time domain reflectometer (TDR)
 - Cable certifiers



Continuity tester



Multimeter



**Medium-priced
TDR**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

84



Mike Meyers' Guide to Managing and Troubleshooting Networks

- Crosstalk
 - **Near-end crosstalk (NEXT) and far-end crosstalk (FEXT)**
 - Depends where the listening and the actual crosstalk exist
- Attenuation/Dispersion
 - **Signal becomes steadily weaker as it progresses down the wire;**
 - **More susceptible to crosstalk**
 - **In fiber optics, the amount of light propagating down the fiber cable diffuses over distance**

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

85



Mike Meyers' Guide to Managing and Troubleshooting Networks

Testing the Cable Runs

Fiber Challenges

- **Potentially**
 - Signal loss
 - **Damaged cables or open connections**
 - **Dirty or mismatched connector**
 - **Fiber length – attenuation/dispersion**
 - **Bend radius – Light leakage occurs with too much bend**
 - Physical or Signal Mismatch
 - **Transceiver, fiber, or wavelength mismatch**
- **Testing tools**
 - Optical Time Domain Reflectometer (OTDR)
 - **Determines continuity**



OTDR



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

86



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Diagnostics and Repair of Physical Cabling

Troubleshooting is an art

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

87



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Diagnostics & Repair

- Visually check that everything is plugged in
- Does the problem affect all computers or only one?
 - If all, the problem is with a shared resource such as the switch or a print server
 - If one, the problem is probably in the host or in the cabling between the host and the switch



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

88



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Diagnostics & Repair

- Is the problem hardware or software?
 - If one application fails, try another
 - If the user can browse the internet, but cannot get his e-mail, it is a software problem not a network issue
 - If nothing can access the network (can't get on the internet, can't print, nothing in "Network Neighborhood") you probably have a hardware problem

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

89



Mike Meyers' Guide to Managing
and Troubleshooting Networks

Use the Lights

- Check if everything is connected & isolate the problem
- UTP NICs have LEDs for state of link
 - Link light steady when NIC is connected
 - Activity light flashes when network traffic is detected
- Many fiber-optic NICs don't have lights
- Switches also have link lights
- **Bonding or link aggregation**
 - Multiple NICs to effectively double speed between device & switch




Copyright © 2022 by McGraw-Hill Education. All rights reserved.

90

Mike Meyers' Guide to Managing and Troubleshooting Networks

- Is the problem in the host or cabling?
 - Plug the system into a known good outlet
 - System works: may be cabling problem
 - Confirm with a continuity test
 - System does not work: may be
 - NIC in the host or a bad connector
 - Incorrect driver installed



Not Connected

You are currently not connected to any networks.



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

91

Mike Meyers' Guide to Managing and Troubleshooting Networks

Cable Testing

- Most problems occur at the work area but...
- To check cables
 - Use tools to check continuity (make sure there are no breaks or disconnections)
 - If there is no continuity, try to isolate the problem to one of the patch cables or the horizontal run

Copyright © 2022 by McGraw-Hill Education. All rights reserved.

92



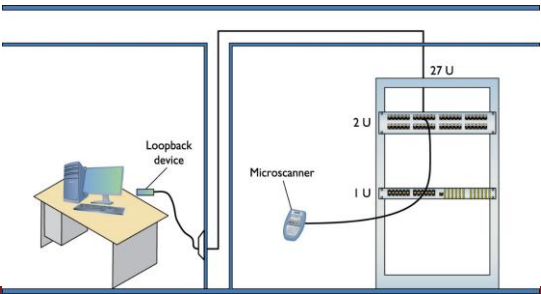
Mike Meyers' Guide to Managing and Troubleshooting Networks

Check the NIC

- Use operating system or available NIC diagnostic software
- Run a **Loopback test on NIC**
 - Internal only checks circuitry
 - External (with **loopback plug**) also tests connector



Loopback plug



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

93



Mike Meyers' Guide to Managing and Troubleshooting Networks

Troubleshoot: Telecom Room

- Keep the diagnostic process organized and documented
 - Start at one end, and don't skip connections
 - Place a sticker as you work to track progress
- Biggest concerns
 - **Uninterrupted power supply (UPS)**
 - Can be a monitoring tool
 - Provides security from power spikes and sags
 - Enables shutdown
 - **Monitoring tools**
 - Voltage event recorder
 - Temperature
 - Environmental



Copyright © 2022 by McGraw-Hill Education. All rights reserved.

94